FD-55A Series

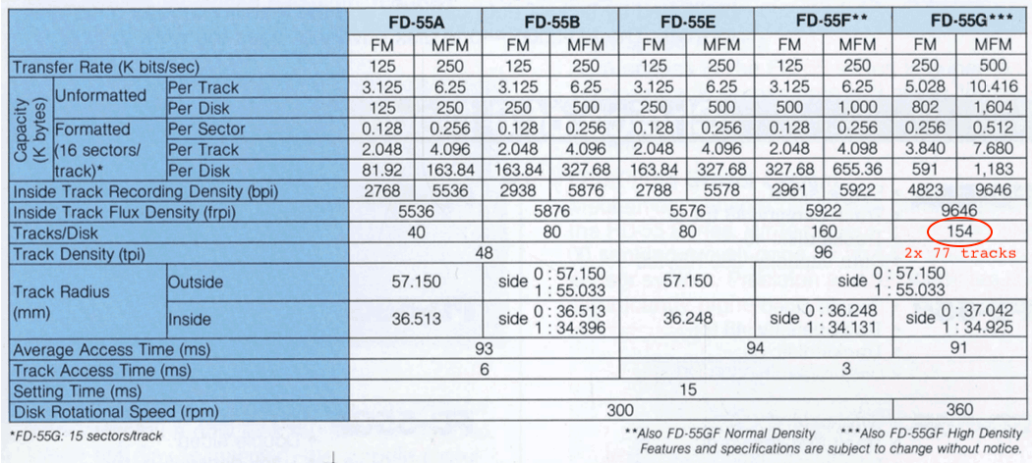
top

- TEAC FD-55A
- TEAC FD-55AV

General

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Technical specifications



			FD-55A		FD-55B		FD-55E		FD-55F**		FD-55G***	
			FM	MFM	FM	MFM	FM	MFM	FM	MFM	FM	MFM
Transfer Rate (K bits/sec)			125	250	125	250	125	250	125	250	250	500
Capacity (K bytes)	Unformatted	Per Track	3.125	6.25	3.125	6.25	3.125	6.25	3.125	6.25	5.028	10.416
		Per Disk	125	250	250	500	250	500	500	1,000	802	1,604
	Formatted (16 sectors/ track)*	Per Sector	0.128	0.256	0.128	0.256	0.128	0.256	0.128	0.256	0.256	0.512
		Per Track	2.048	4.096	2.048	4.096	2.048	4.096	2.048	4.098	3.840	7.680
		Per Disk	81.92	163.84	163.84	327.68	163.84	327.68	327.68	655.36	591	1,183
Inside Track Recording Density (bpi)			2768	5536	2938	5876	2788	5578	2961	5922	4823	9646
Inside Track Flux Density (frpi)			5536		5876		5576		5922		9646	
Tracks/Disk			40		80		80		160		154	
Track Density (tpi)			48				96				2x 77 tracks	
Track Radius (mm)	Outside		57.150		side 0 : 57.150 1 : 55.033		57.150		side 0 : 57.150 1 : 55.033			
	Inside		36.513		side 0 : 36.513 1 : 34.396		36.248		side 0 : 36.248 1 : 34.131		side 0 : 37.042 1 : 34.925	
Average Access Time (ms)			93				94				91	
Track Access Time (ms)			6				3					
Setting Time (ms)			15									
Disk Rotational Speed (rpm)			300								360	

*FD-55G: 15 sectors/track

**Also FD-55GF Normal Density ***Also FD-55GF High Density
Features and specifications are subject to change without notice.

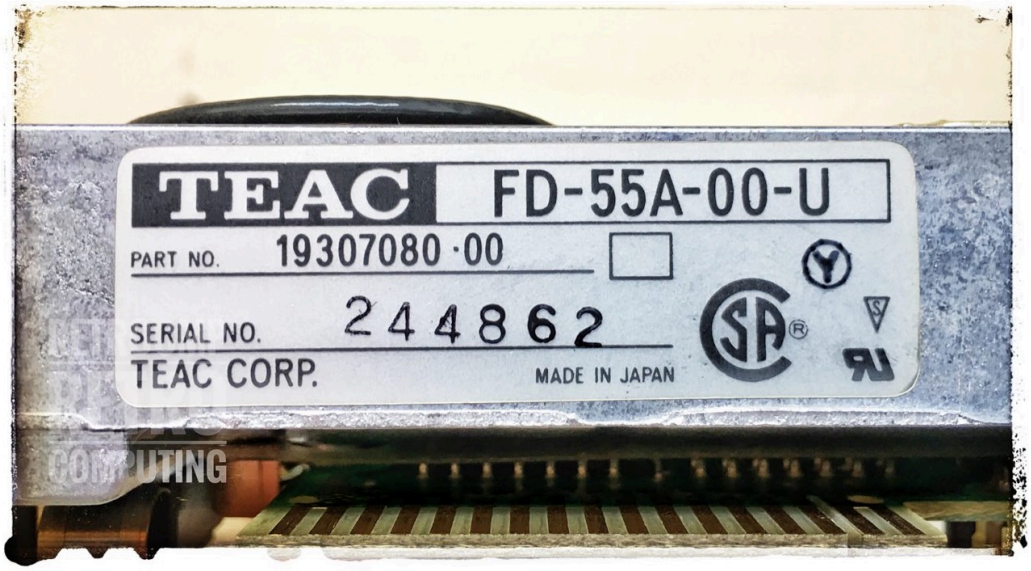
TEAC FD55 technical specifications
(Source: Brochure TEAC FD-55 Series)

The TEAC FD-55A is the first model of the 55 series and was actually only used in the early days of CP/M and DOS, because it can only read/write on one side. Already from PC-DOS 1.1 double sided disks were supported.

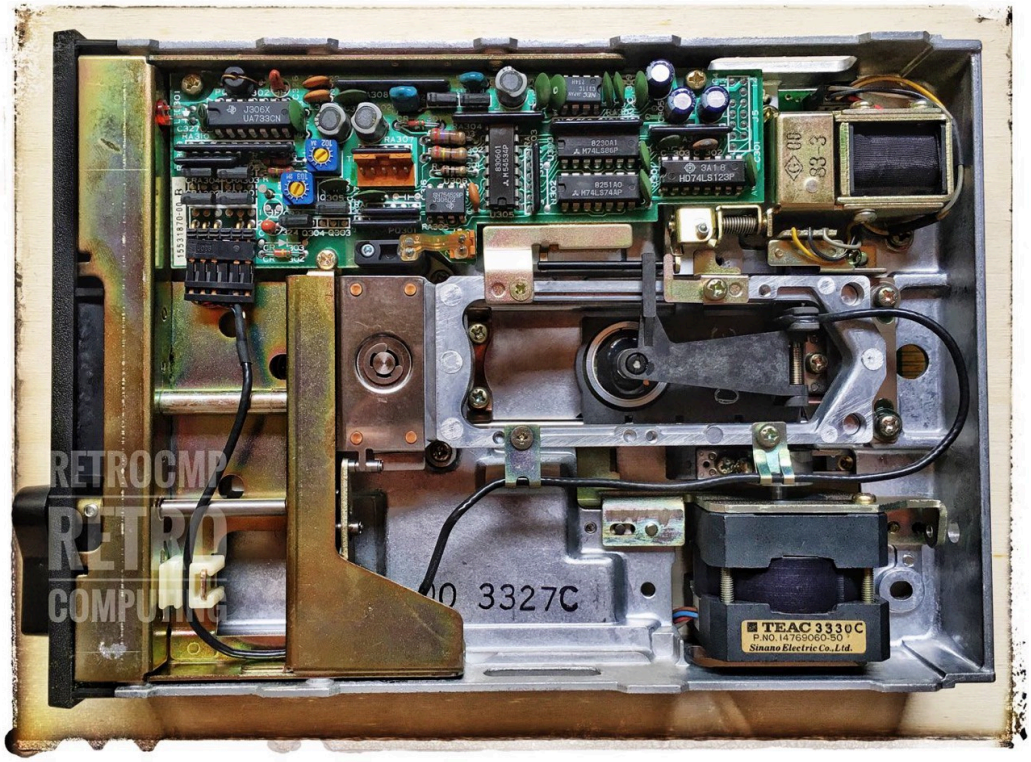
Like many of these very first drives, the TEAC FD-55A also has a real head loading magnet. My drive looks brand-new. I don't think it's ever been used.

FD-55A

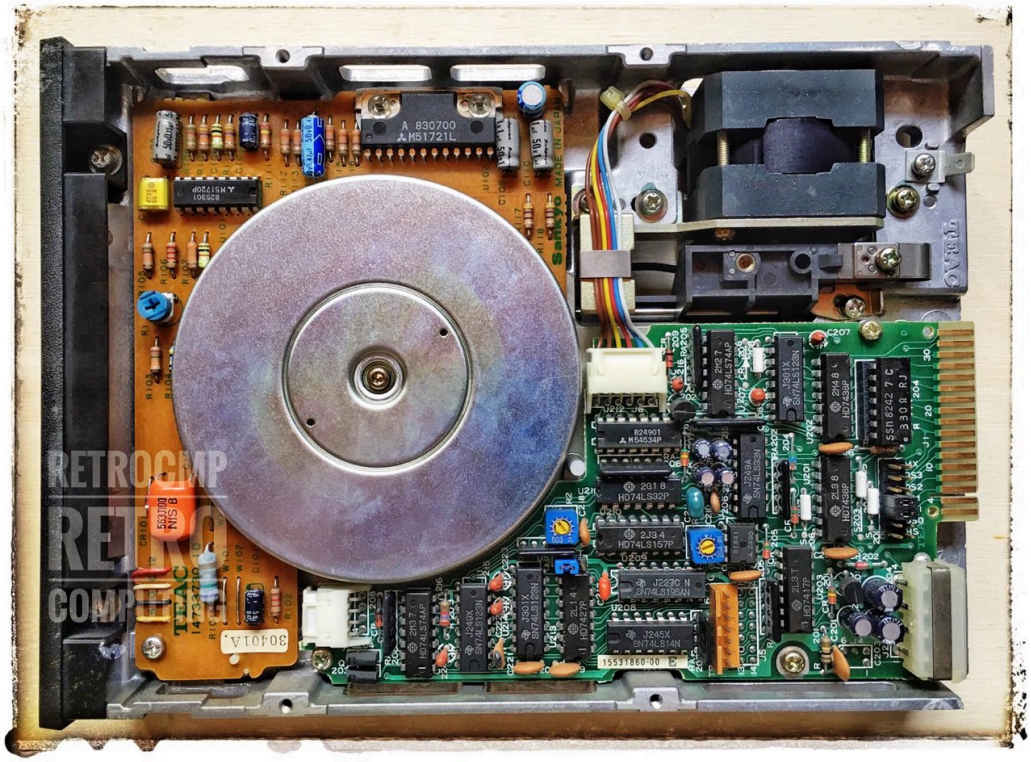
top



(1) TEAC FD-55A (1930-7080-00)



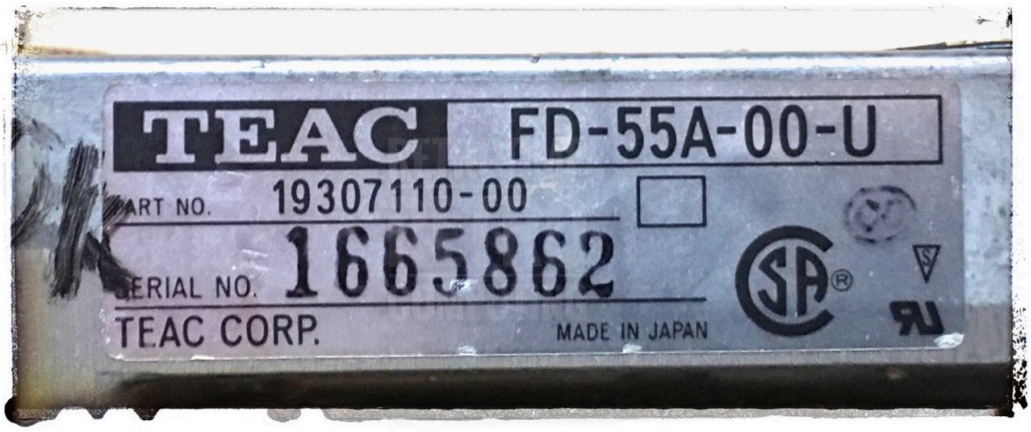
(1) TEAC FD-55A (1930-7080-00)



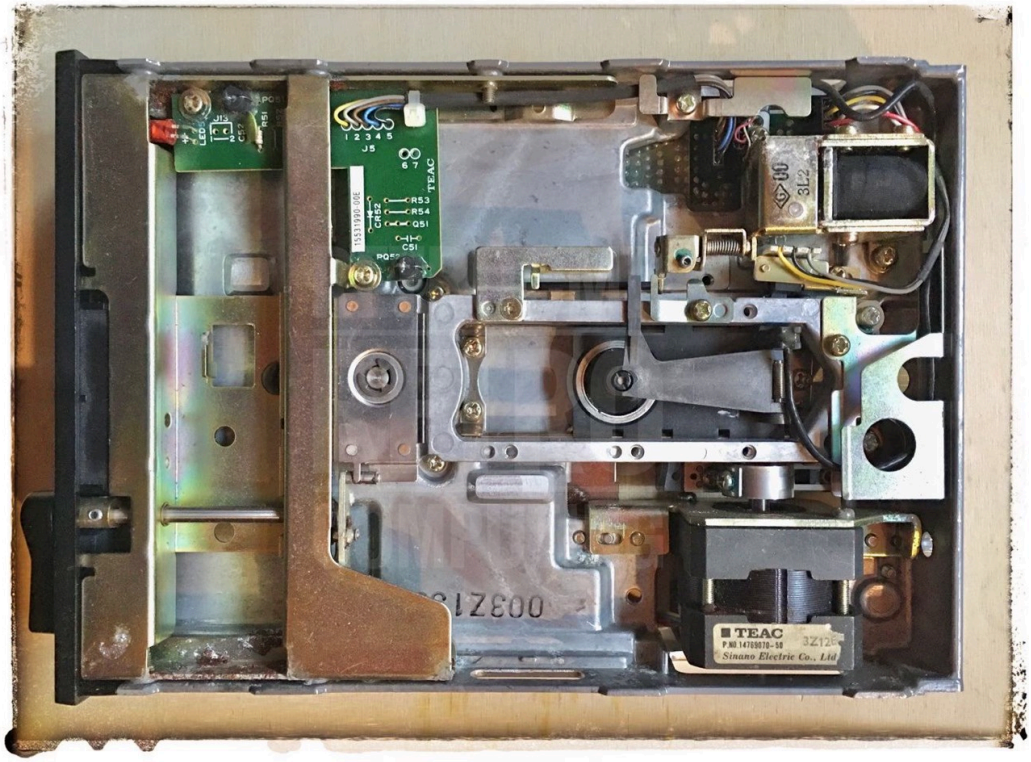
(1) TEAC FD-55A (1930-7080-00)
non LSI, 330 Ohm terminator

My 1st TEAC FD-55A-00-U (part no. 1930-7080-00) is an early TTL version of the A series. I have a lot of TEAC drives, but this is one of the few *TTL* versions I have. All others are LSI.

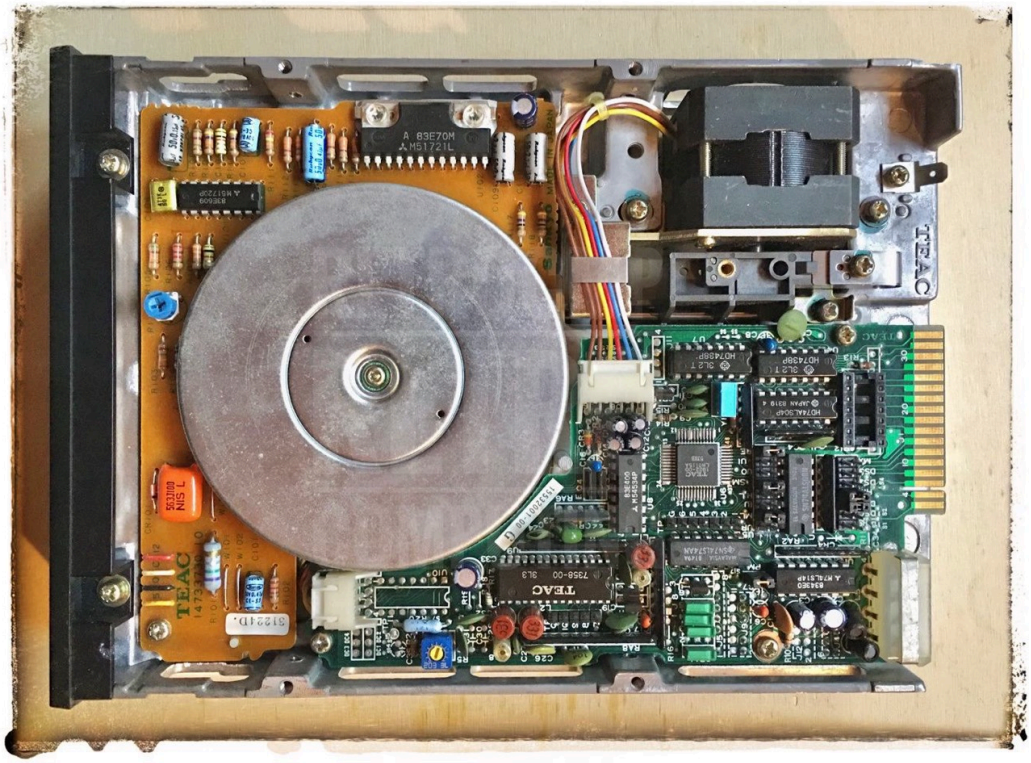
Another special feature: I got this drive with a 330 Ohm terminator pack, not the usual 150 Ohm resistor! See figure above.



(2) TEAC FD-55A (1930-7110-00)

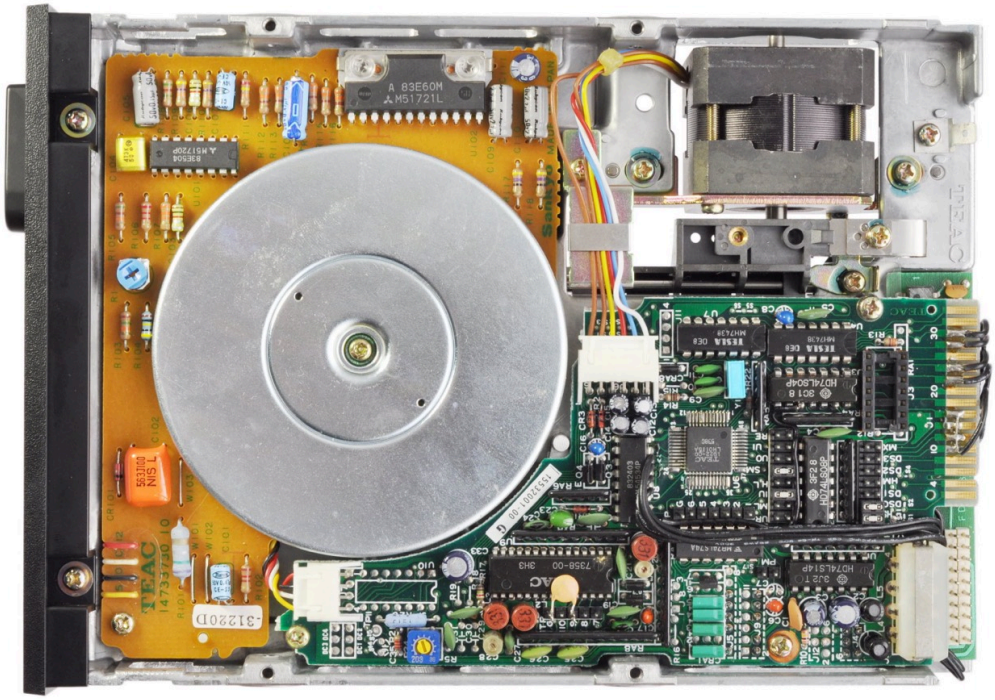


(2) TEAC FD-55A (1930-7110-00)

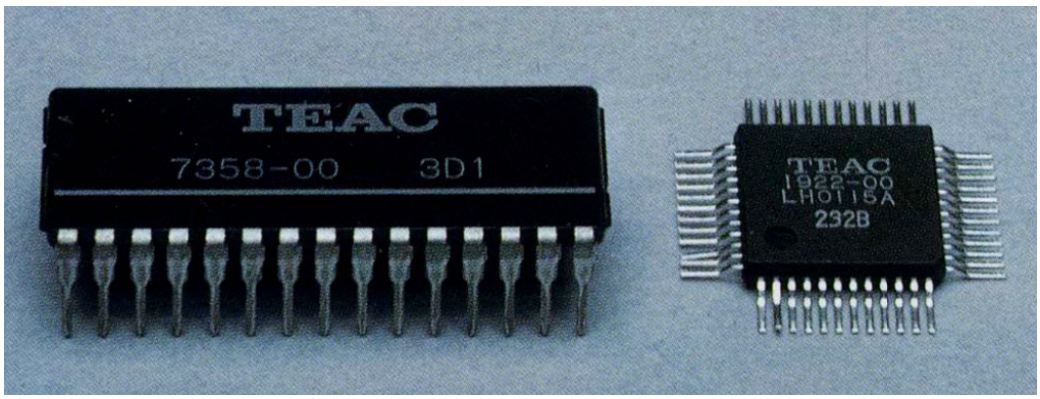


(2) TEAC FD-55A (1930-7110-00)
LSI, 150 Ohm terminator

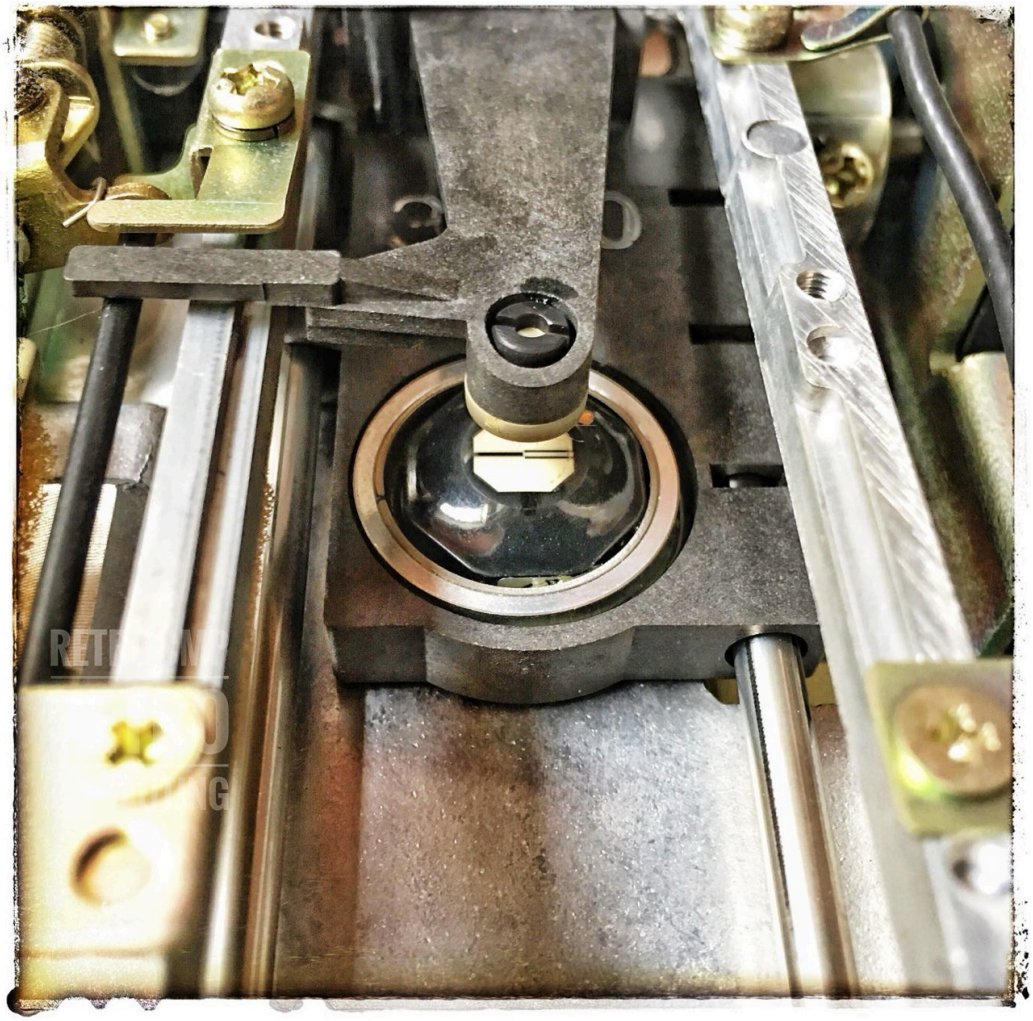
My 2nd TEAC FD-55A-00-U (part no. 1930-7110-00) is newer *LSI* version of the A series. The PCB match the one in the brochure (1985) below.



TEAC FD-55A (Vogons Wiki)



TEAC LSIs for the newer FD55 series



Button type read/write head with tunnel erase

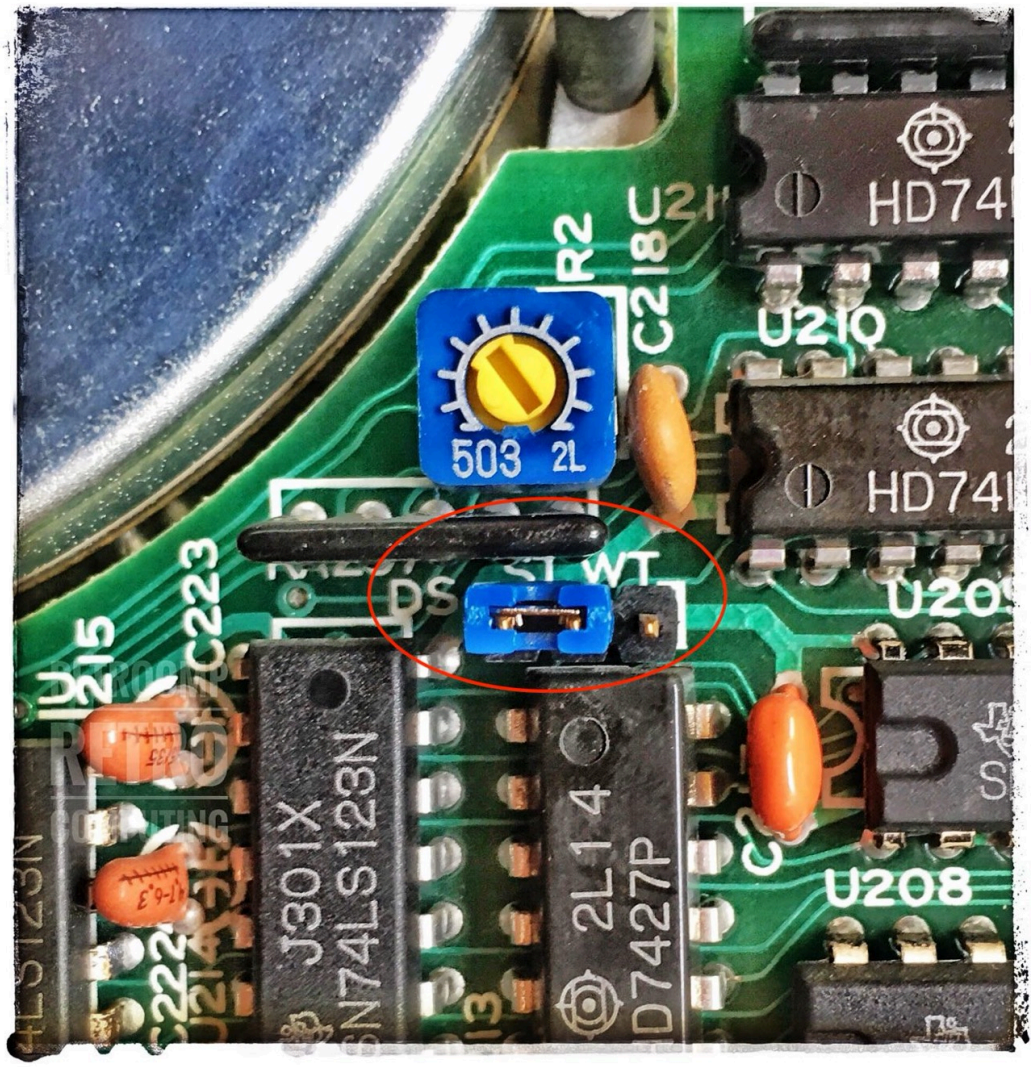
> *The head load mechanism functions to make the head in contact with the disk when only required so as to reduce disk and magnetic head wear. This mechanism is constructed with head load solenoid, arm lifter, etc. For a single sided FDD, the head pad attached to the pad arm of the head carriage is depressed against the SIDE 1 of the disk with appropriate pressure and the disk is held with head and head pad.*
> *TEAC FD55 Maintenance Manual, p. 304*

Jumper settings

The FD-55A has no DISK CHANGE (DC) and no READY (RY) jumper. In other words, this drive does not work properly on a newer AT, only on an older PC/XT.

There are actually only the jumpers HS, DS0, DS1, HM, DS2, DS3 and MX. My jumper settings are: DS1 and HM. That's all. But ... There is a strange jumper called DS with the settings ST or WT.

Also with the jumpers it is noticeable that this drive/PCB is different. A complete row of jumpers is missing (see: TEAC FD-55 Drive Specification, p. 128).



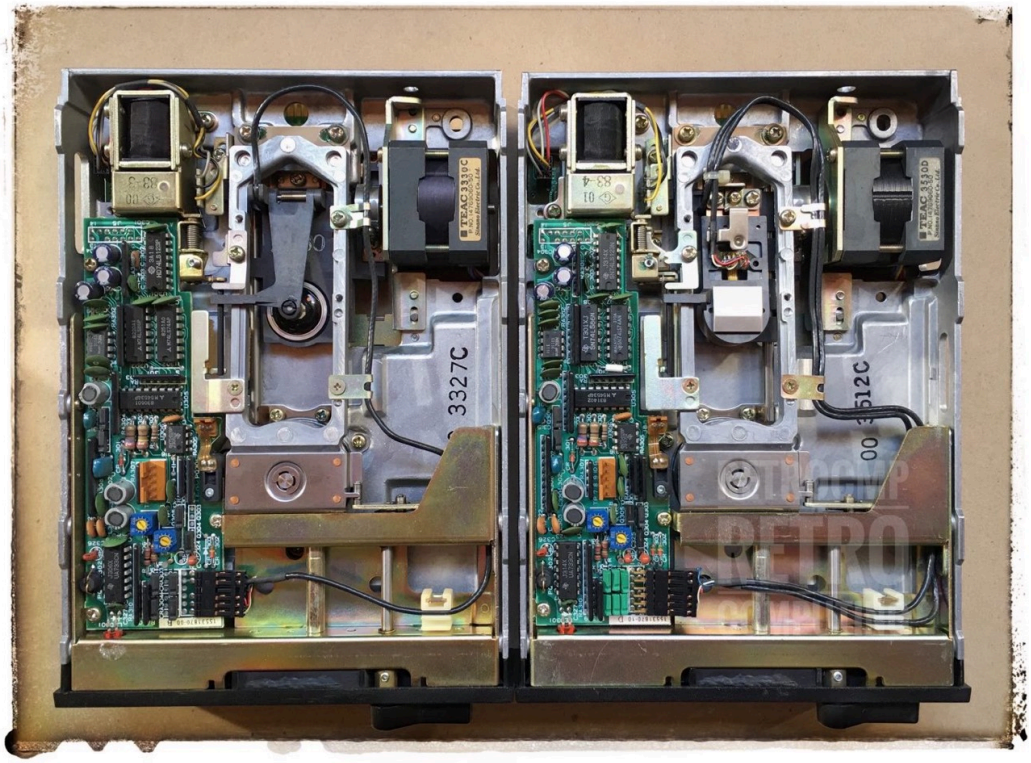
A strange jumper named DS (ST/WT)

You have to plug the jumper on ST. If you do not, the stepper uses 80 tracks and the floppy is not usable.

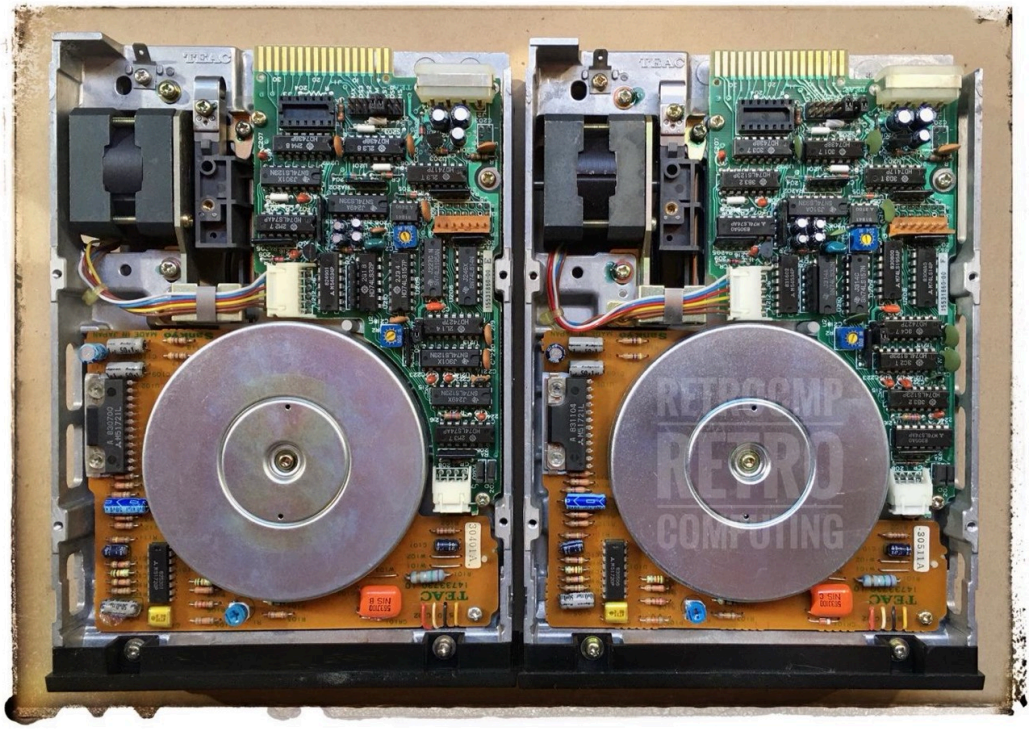
With a previously formatted diskette, the table of contents can be read with DIR, but all other file accesses are incorrect. This old and wide „button type" r/w head can process 80 tracks but they are overlapping. Only the newer and „narrower" r/w heads of the 96 tpi drives can do this!

> *Chuck(G), 25.04.2019, VCF*
> *TEAC used the same stepper on all of its 40 and 80 track drives. The 55A uses the same PCB as the 55E (80T, SS), so the jumper is useful.*

Update 25.08.2020: TEAC uses this PCB also with the 55B, see next figures.



TEAC FD-55A-00-U / TEAC FD-55B-01-U



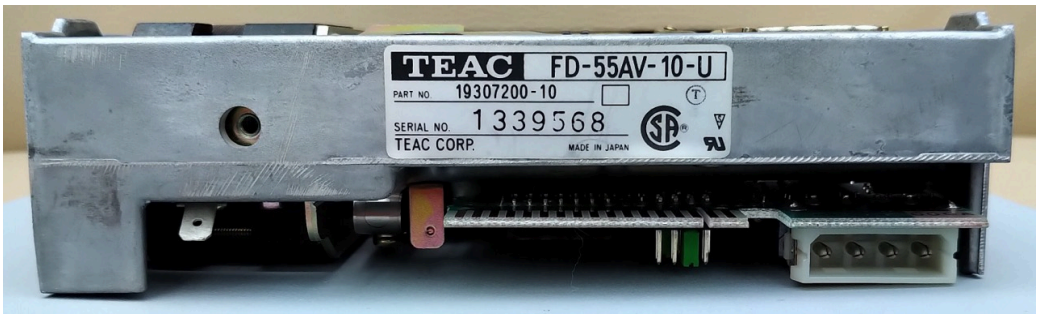
TEAC FD-55A-00-U / TEAC FD-55B-01-U

FD-55AV

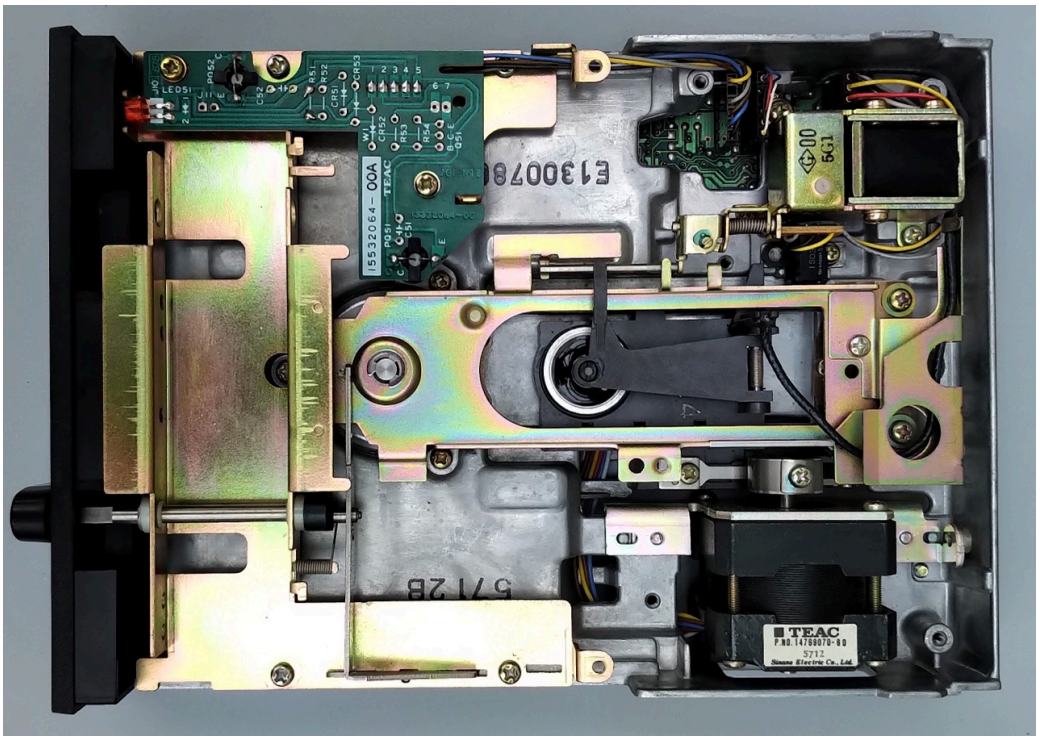
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General

Here are a few more pictures of the TEAC FD-55AV. These were kindly provided by *as58* from VzEkC.



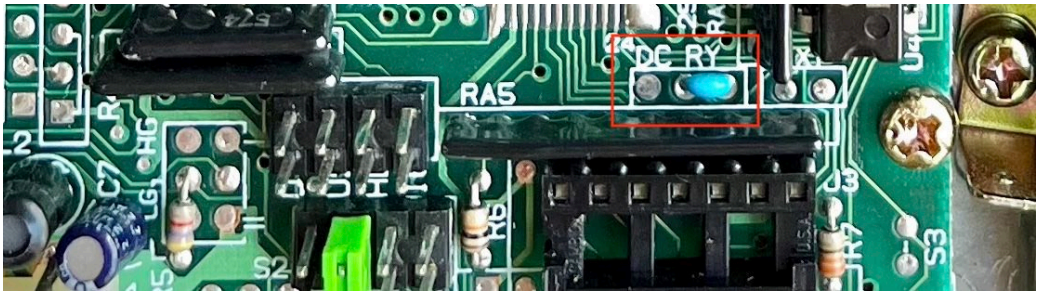
TEAC FD-55AV-10-U (by as58 from VzEkC)



TEAC FD-55AV-10-U (by as58 from VzEkC)

A comparison of the FD-55AV boards with the FD-55A (non LSI) shows very nicely the further technical development.

In contrast to the FD-55A the AV version has solder pads for RY or DC; see images. However, these are not constructed as a jumper.



TEAC FD-55AV-10-U with RY/DC solder pads



TEAC FD-55AV-10-U (by as58 from VzEkC)

Maintenance

I was able to buy a FD-55AV cheaply in 2022. Basically it works, but only with its own formatted floppy disks. This implies that there is probably a *track 00* problem.

... to be continued ... There is a very detailed maintenance manual from the 55AV.

Specifications (S) and brochures (B)

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- S: TEAC FD-55AV
- S: TEAC FD-55 Jumper Settings
- B: TEAC FD-55 Series (A, B, E, F, G, GF)

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